

Absolute Auditory Thresholds in Three Old World Monkey Species (*Cercopithecus aethiops*, *C. neglectus*, *Macaca fuscata*) and Humans (*Homo sapiens*)

Michael J. Owren, Steven L. Hopp, Joan M. Sinnott, and Michael R. Petersen
Indiana University

We investigated the absolute auditory sensitivities of three monkey species (*Cercopithecus aethiops*, *C. neglectus*, and *Macaca fuscata*) and humans (*Homo sapiens*). Results indicated that species-typical variation exists in these primates. Vervets, which have the smallest interaural distance of the species that we tested, exhibited the greatest high-frequency sensitivity. This result is consistent with Masterton, Heffner, and Ravizza's (1969) observations that head size and high-frequency acuity are inversely correlated in mammals. Vervets were also the most sensitive in the middle frequency range. Furthermore, we found that de Brazza's monkeys, though they produce a specialized, low-pitched boom call, did not show the enhanced low-frequency sensitivity that Brown and Waser (1984) showed for blue monkeys (*C. mitis*), a species with a similar sound. This discrepancy may be related to differences in the acoustics of the respective habitats of these animals or in the way their boom calls are used. The acuity of Japanese monkeys was found to closely resemble that of rhesus macaques (*M. mulatta*) that were tested in previous studies. Finally, humans tested in the same apparatus exhibited normative sensitivities. These subjects responded more readily to low frequencies than did the monkeys but rapidly became less sensitive in the high ranges.

This study was part of a project examining basic hearing functions in three Old World primate species: vervet monkeys (*Cercopithecus aethiops*), de Brazza's monkeys (*C. neglectus*), and Japanese macaques (*Macaca fuscata*). Elsewhere we have described our investigations of other aspects of hearing in these animals—sensitivity to differences in frequency, intensity, and duration (Sinnott, Owren, & Petersen, 1987a, 1987b; Sinnott, Petersen, & Hopp, 1985); acuity for aspects of human speech sounds (Sinnott & Adams, 1987); and processing of species-typical vocalizations (Hopp, 1985; Owren, 1986; Petersen, 1982). This article reports absolute auditory thresholds, examines the possibility of species differences, and allows more precise specification of sensation levels that monkeys experience when listening to auditory stimuli.

Early reviews of primate hearing did not report any significant differences in absolute sensitivity among Old World monkeys (e.g., Stebbins, 1973). However, two more recent studies indicated that variation may exist. Hienz, Turkhan, and Harris (1982) found that yellow baboons (*Papio cynoce-*

phalus) are consistently about 5 dB more sensitive across the audible frequency range than are macaques that were tested in previous studies. In addition, Brown and Waser (1984) found low-frequency sensitivity in blue monkeys (*C. mitis*) superior, relative to the best results previously reported for rhesus monkeys (*M. mulatta*). However, the subjects in both of these studies were tested in a free-field setting, and some of the heightened sensitivity reported may be attributed to this fact. The investigations that Stebbins reviewed had used closely fitting earphones, and it is difficult to make comparisons across these two acoustic environments (Killion, 1978; Rudmose, 1982). In our study we avoided this problem by using identical testing procedures for all subjects.

The species were selected for our study because, though all three are Old World monkeys, they exhibit substantial variation in vocal behavior, socioecology, and morphology. Differences in hearing may therefore also be expected. For example, Masterton, Heffner, and Ravizza (1969) proposed that mammals with small interaural distances (i.e., small heads) tend to rely on acute high-frequency sensitivity for sound localization (see R. S. Heffner & Heffner, 1982, for a recent review). They pointed out that though animals with relatively large interaural distances can respond to disparities in sound arrival time at each ear, smaller species may more easily exploit the effects of acoustical shadowing by the head. Increases in the upper frequency limit of hearing may allow significant shadowing-related alterations of the frequency-intensity spectrum of a given signal even in animals that are quite small. Although normative adult interaural distances are not available for our subjects, these species differ in weight, overall size, and general cranial dimensions (Hill, 1966, 1971). Japanese macaques are the largest of the three, followed by de Brazza's monkeys, and vervets are clearly the smallest.

Portions of these data were presented at the Acoustical Society of America meeting, Chicago, Illinois, 1982, and the Animal Behavior Society regional meeting, Champaign, Illinois, 1982. The research was supported by grants from the National Science Foundation (BNS 79-24477), the Deafness Research Foundation, the Spencer Foundation, and the Alfred P. Sloan Foundation to Michael R. Petersen.

We thank Charles Brown, Robert Seyfarth, and an anonymous reviewer for comments and suggestions on earlier versions of this article; Roderick Suthers and Donald Robinson for helpful discussions; and Susan Woontner both for participating as a subject and for testing monkeys.

Correspondence concerning this article should be addressed to Michael J. Owren, California Primate Research Center, University of California, Davis, California 95616.

The superior low-frequency sensitivity reported for blue monkeys may also be expected for de Brazza's monkeys. Both are forest-dwelling guenons that produce an unusual, loud, low-pitched boom call (Brown & Waser, 1984; Gautier-Hion & Gautier, 1978). Brown and Waser specifically associated the enhanced low-frequency sensitivity that blue monkeys exhibit with the use of this call for long-distance signaling between neighboring groups.

Japanese macaques were of great interest in the larger context of our studies because of the reported humanlike perceptual strategies of these animals when they listen to species-typical vocal signals (H. E. Heffner & Heffner, 1984; Hopp, 1985; Petersen, 1982; Petersen et al., 1984). In addition, inclusion of these monkeys allowed a partial check of our findings through comparison with previous reports. The hearing of *Macaca* species is among the best studied of that of any Old World monkey genus. Other investigators have studied several macaques by using procedures similar to ours (e.g., Pfingst, Laycock, Flammino, & Lonsbury-Martin, 1978; Stebbins, Green, & Miller, 1966).

Method

Subjects

The subjects were 3 vervets, 5 de Brazza's monkeys, 5 Japanese macaques, and 4 humans. All the monkeys were juveniles or subadults when testing began, except for 2 vervets and a de Brazza's monkey that were young adults. One de Brazza's monkey was a female. The vervets weighed 2.6 to 4.2 kg, the de Brazza's monkeys 2.5 to 7.0 kg, and the Japanese macaques 3.2 to 5.5 kg. All the monkeys were experimentally naive prior to training. They were housed in individual cages with free access to water and maintained on a 22.5-hr restricted-feeding schedule. They earned a large proportion of their food ration during the daily testing sessions and received the remainder in the form of Purina High-Protein Monkey Chow and fresh fruit immediately after each session. On alternate days the animals were also given a chewable multivitamin supplement.

The human subjects were three of the authors (1 female and 2 males; ages 33, 24, and 25) and a research assistant (female, age 20). One subject was an experienced sensory observer at the start of the study, whereas the others had not previously participated in rigorous psychophysical testing. None of the subjects had a known history of acoustical or hearing-related trauma, although one subject was believed to have suffered a minor high-frequency-hearing loss.

Apparatus

The apparatus has been described in detail elsewhere (Sinnott et al., 1987b; Sinnott et al., 1985). Testing was conducted in a single-walled, sound-attenuating booth. The monkeys sat in a primate restraint chair, and the humans sat in an ordinary straight-backed chair. The subjects responded by using one of three touch-sensitive, stainless steel tubes attached to an intelligence panel directly in front of them. The tubes were mounted at eye level to the monkeys, each tube containing a 12-V neon lamp. The monkey's head was restrained by a Plexiglas plate at the back, ear cushions, and a brass rod on either side of the snout. A copper tube for delivering 190 mg banana-flavored food pellets (Noyes Formula G) was mounted near the monkey's mouth, and pellets were dispensed from a feeder located above the intelligence panel. All experimental procedures were controlled and monitored with a DEC PDP-11/34 computer system.

Sinusoidal pure-tone stimuli were generated by a Krohnkite (Model 4141R) programmable oscillator. Signals passed through an electronic switch with rise/fall times of 10 ms to a custom-designed and fabricated programmable attenuator and then through a manual attenuator network. The tones were subsequently fed to a Crown amplifier (Model D75) and went from there to a pair of TDH 49P earphones that were centered carefully over the subject's pinnae. To eliminate signal artifacts and harmonic distortion at low (0.063–0.250 kHz) and high (16.0–45.0 kHz) frequencies, a Krohnkite (Model 3550R) band-pass filter was inserted into the signal-generating circuit and set to $\frac{1}{4}$ octave above and below the stimulus frequency. Signal characteristics were monitored with an analog voltmeter and a digital frequency counter.

For frequencies from 0.063 to 45.0 kHz, the intensity of the signal produced by each earphone was initially calibrated in a closed system with a $\frac{1}{2}$ -in. (1.5 cm) Bruel and Kjaer (Type 4134) condenser microphone inserted into a 6-cc NBS (Type 9-A) earphone coupler. The microphone output was passed to a General Radio (Model 1900-A) wave analyzer that permitted measurements of signal level and harmonic distortion. The most prominent harmonics measured were 40 to 50 dB lower in amplitude than the fundamental frequency. For frequencies from 0.063 to 4.0 kHz, the signal level was also measured with a Knowles (Model 1843) miniature electret condenser microphone placed at the entrance to the external auditory canal of a ketamine-anesthetized monkey (one monkey of each species was tested) or an awake human.

There were no differences between the sound pressure levels measured with the earphone coupler system and with the electret microphone when worn by a human. For the monkeys signals from 0.063 to 2.0 kHz yielded large (5–15 dB) and reliable differences between these two systems. However, less than a 2-dB variation was found between monkey species when each calibration technique was considered separately. This outcome suggests that the 6-cc earphone coupler, the dimensions of which approximate the volume enclosed within the human outer and middle ear when an earphone is worn, is inappropriate for calibrations of the monkey ear's response to low-frequency signals. (See Table A-1 in the Appendix for comparisons of these two methods of calibration.) Therefore, for all monkeys measurements of signal levels from 0.063 to 2.0 kHz were taken from the electret microphone system. (Its response falls off so quickly beyond 4.0 kHz that it is useless above this frequency.) Those between 2.8 and 45.0 kHz were taken from the condenser microphone system.

Procedure

The auditory-testing procedure was similar for monkeys and humans. A trial sequence began with on-off flashing of the cue light inside the response tube. When the subject made contact with the tube, the cue light remained on steadily, and a randomly variable hold of 0.5 to 8.0 s was initiated. At the end of 85% of the holds, a 1.5-s tone stimulus was presented. A release of the tube during the tone extinguished the cue light, terminated the tone, and produced a reward (a food pellet for the monkeys and the click of the empty dispenser for the humans).

After a reward the cue light was extinguished, and the subject entered an intertrial interval (ITI). This period was adjusted in length from 5.0 to 12.0 s for individual monkeys, depending on how quickly they chewed the pellet. The ITI was fixed at 1.0 s for the humans. Any tube contact during the ITI started it anew. Failure to release the tube during the tone resulted in the initiation of a new variable hold without feedback to the subject.

At the end of 15% of all holds, a 1.5-s catch trial was presented to measure the false-alarm (guessing) rate of each subject. If the subject maintained contact with the tube during the catch trial (i.e., correct rejection), a new variable hold was started without feedback. How-

ever, tube releases occurring during a catch trial or a variable hold (i.e., false alarms) extinguished the cue light and resulted in 7.0-s time-out from the procedure. Any tube contact during a time-out reset the interval. For monkeys a session ended when the subject earned its daily allotment of food pellets; depending on the subject's weight and appetite, the total number of pellets earned ranged from 100 to 200. Human sessions terminated after 100 correct detections.

Tones at $\frac{1}{2}$ -octave intervals from 0.063 to 45.0 kHz were used in testing. Additional frequencies between 32.0 and 45.0 kHz (34.9, 38.0, 40.0, 42.0, and 43.0) were used to help define the shape of the audiogram at higher frequencies. All stimuli were presented binaurally.

We divided the sessions into two to four blocks of trials and tested a different frequency during each block. For a given block seven equally spaced tone intensities, spanning a 14–35 dB range, were randomly selected according to the method of constant stimuli. Psychometric functions were constructed by plotting the percentage of tone detections (correct release responses) as a function of sound-pressure level (SPL). Each data point in these functions represented at least 10 trials. The threshold was defined as the 50% level of detection. Data from blocks of a session in which the catch trial response exceeded 15% were discarded. Testing at each frequency continued until the thresholds collected from an individual over 15 consecutive sessions fell within a 10-dB range and there was no further downward trend in the thresholds. The mean of the five lowest threshold measurements made during this period was calculated and used in the final audiograms.

Results

In Figure 1 we present the psychometric functions that were obtained from one of the Japanese monkeys at four frequencies. The form of the functions, especially with respect to their ogival shape and steep slopes, is representative of those obtained from all three monkey species as well as the humans tested. Although some animals produced higher false-alarm rates than others, the species did not differ overall.

Mean thresholds for all species at frequencies up to 32.0 kHz are shown in Table 1. Table 2 shows analysis of variance results for each frequency at which significant differences were obtained, both with and without human subjects. The vervets provided reliable data at all frequencies. However, at 42.0 kHz and above, the de Brazza's monkeys and Japanese macaques became increasingly unable to respond to even the loudest signals that we could produce (about 72 dB SPL). We give the thresholds for the highest frequency region in Table 3, which includes only reliably responding individuals.

The mean thresholds for each monkey species and 2 humans appear in Figure 2. We found that the 2 human subjects that are not shown in this figure were consistently less sensitive to frequencies between 5.6 and 16.0 kHz than were the others. These 2 were the oldest subject and the subject with a suspected high-frequency loss.

In the low-frequency region humans were significantly more sensitive than were the monkeys at 0.250, 0.353, and 0.500 kHz. However, the monkey species did not differ from each other. Sensitivity to low frequencies diminished at about the same rate for all species; that is, each audiogram shows a roll-off of approximately 18 dB per octave for frequencies below 1 kHz. In the middle frequency region (1.0–4.0 kHz) the vervets appeared to be more sensitive than the other species, and this difference was significant at 2.0 kHz.

At high frequencies the humans' sensitivity decreased rapidly (and significantly) beyond 8.0 kHz, falling off at a rate of 27 dB per octave. The monkeys' hearing extended well beyond this point, up to 45.0 kHz in the case of the vervets. Figure 3 highlights the performance of the reliably responding animals at the highest frequencies tested. Although the hearing of all three monkey species declined at a rate of roughly 160 dB per octave above 34.8 kHz, the vervets were consistently more sensitive than the others. The differences generally increased with frequency, until only the vervets were able to respond.

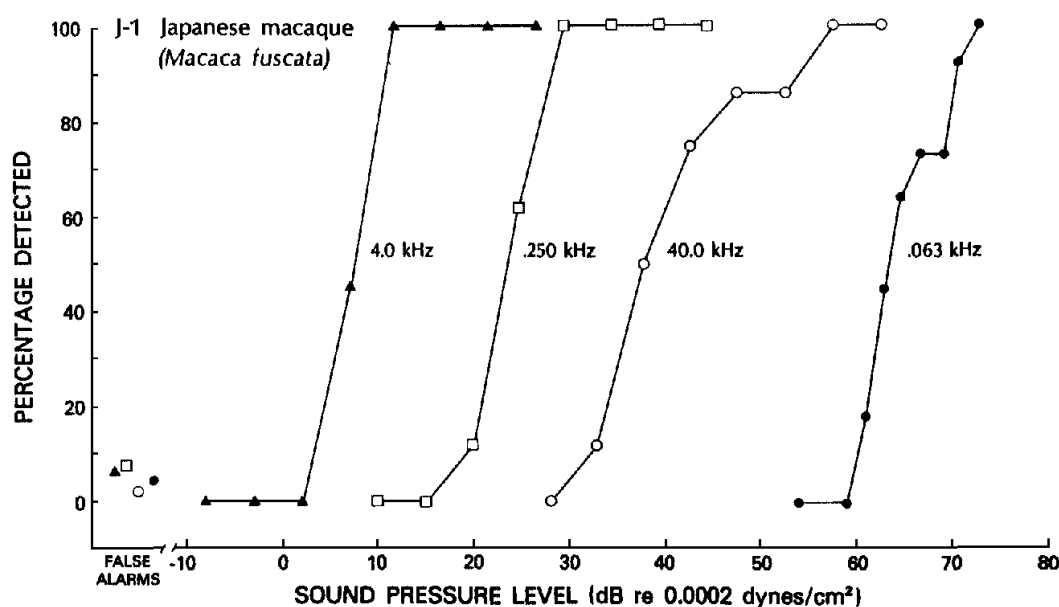


Figure 1. Psychometric functions obtained from one Japanese macaque subject at four frequencies.

Table 1

Mean Thresholds, Standard Deviations, and Ranges (in Decibels) for All Subjects at Frequencies up to 32.0 kHz

Frequency (in kilohertz)	<i>Cercopithecus aethiops</i> (n = 3)			<i>Cercopithecus neglectus</i> (n = 5)			<i>Macaca fuscata</i> (n = 5)			Humans (n = 4)		
	M	SD	Range	M	SD	Range	M	SD	Range	M	SD	Range
0.063	62	5.0	57-67	59	4.8	56-66	68	4.7	64-74	61	4.5	55-65
0.088	54	4.0	52-59	55	8.6	42-64	58	6.9	49-65	53	3.2	50-57
0.125	42	0.6	42-43	43	5.6	36-49	48	6.9	37-55	42	2.1	39-44
0.176	34	3.8	32-38	34	6.2	29-43	38	7.5	28-47	33	6.7	24-40
0.250	26	3.1	23-29	27	4.0	23-32	32	6.9	26-43	21	3.6	16-24
0.353	16	2.6	14-19	16	1.3	15-18	21	4.5	14-26	11	6.0	4-18
0.500	8	3.8	7-10	11	2.3	8-13	15	4.9	7-20	6	2.9	4-10
0.707	3	2.1	1-5	7	4.0	3-14	9	4.4	2-14	2	5.5	-5-8
1.000	-2	3.1	-3-1	4	1.6	2-6	3	4.6	-2-10	2	1.3	0-3
1.414	-4	2.6	-6--2	3	3.0	1-8	3	5.5	-3-11	7	2.9	4-11
2.000	6	1.9	5-8	15	2.8	13-18	11	3.8	6-16	11	4.9	7-18
2.828	8	3.2	4-10	12	2.3	9-14	15	4.8	10-22	13	8.3	1-19
4.000	3	4.0	-1-7	9	3.3	4-12	10	3.9	7-16	15	7.2	4-19
5.657	-2	6.0	-8-4	2	1.8	0-4	2	5.3	-2-8	13	7.7	4-22
8.000	5	8.1	-1-14	4	3.4	-2-6	8	2.1	6-11	16	6.8	6-22
11.313	4	0.0	4-4	7	2.7	3-12	10	3.1	6-13	38	10.9	29-50
16.000	12	10.0	1-21	6	3.8	3-12	9	4.2	4-14	35 ^a	35.5 ^a	26-45 ^a
22.625	16	5.7	11-22	18	4.7	14-26	18	2.4	15-21	49 ^b	—	—
32.000	15	8.9	5-22	22	6.2	12-28	21	6.9	12-25	—	—	—

^a Based on two humans. ^b Based on one human.

Discussion

Comparisons With Other Psychoacoustic Data

Our data can be compared with previous results from humans (see Figure 4; Dadson & King, 1952; Pfingst, Hienz & Miller, 1975; Sivian & White, 1933) and from several other macaque species (see Figure 5; Pfingst et al., 1978; Stebbins et al., 1966). All thresholds in these figures were measured with earphones, and the data from the various studies are in good agreement (i.e., generally less than 10-dB variability).

The human audiogram presented in Figure 4 is based on results from our 2 most sensitive high-frequency listeners. The only notable discrepancy that appears between our results and those of other studies is that our subjects were 10 to 20 dB more sensitive at 16.0 kHz. In all likelihood this outcome

reflects our deliberate selection of the 2 subjects with the most acute high-frequency hearing.

Disparities of more than 10 dB are also apparent in the macaque results at 0.125 and 0.250 kHz (Figure 5). In this study our Japanese monkeys appear to be less sensitive than any of the other species. However, this may stem from a difference in the low-frequency sound-attenuating characteristics of the testing chambers used in the different studies. Both Pfingst et al. (1978) and Stebbins et al. (1966) conducted their experiments in double-walled chambers that are much more effective in reducing ambient low-frequency noise levels than the single-walled booth used in the present study.

The data shown in Figure 6 allowed us to test this hypothesis. In this figure the sensitivity functions from our animals are compared with other studies by normalizing results with respect to human hearing. Both Brown and Waser (1984) and Pfingst et al. (1975) tested human subjects in the environment used for their animals (blue monkeys and rhesus macaques, respectively). Figure 6 shows the results of plotting the sensitivity of all these primate species relative to the human thresholds measured in each study. Points that lie above 0 dB indicate that the monkey was less sensitive than the average human control; points below 0 dB indicate that the monkey was more sensitive.

In an examination of the relative thresholds of rhesus and Japanese macaques, the normalized data reveal similar sensitivity to stimuli between 0.125 and 0.500 kHz. Moreover, both species are approximately 10 dB less sensitive than humans to these frequencies. Behar, Cronholm, and Loeb (1965) reported the same results at 0.050, 0.100, and 0.125 kHz for rhesus monkeys and humans tested in a free-field environment, but in their experiment animals performed under shock-avoidance contingencies. Thus, the elevated thresholds of our macaques apparent in Figure 5 at 0.125 and

Table 2

Analysis of Variance Results for Frequencies up to 32.0 kHz at Which Significant Differences Occurred, Analyzed Both With and Without Human Subjects

Frequency (in kilohertz)	No humans		Two humans		All humans	
	F(2, 11)	p	F(3, 12)	p	F(3, 14)	p
0.250	1.37	.293	3.43	.052	3.77	.036
0.353	2.88	.090	6.52	.007	5.21	.013
0.500	1.71	.225	2.81	.085	3.65	.039
2.000	4.62	.035	2.73	.090	2.57	.096
5.657	0.28	.972	0.50	.689	3.35	.050
8.000	1.09	.369	1.17	.363	3.73	.037
11.313	1.32	.305	23.31	.000	—	—
16.000	1.17	.345	10.12	.001	—	—

Table 3

Mean Thresholds, Standard Deviations, and Ranges (in Decibels) for Monkey Subjects at Frequencies Above 32.0 kHz

Frequency (in kilohertz)	<i>Cercopithecus aethiops</i>				<i>Cercopithecus neglectus</i>				<i>Macaca fuscata</i>			
	<i>M</i>	<i>SD</i>	Range	<i>n</i>	<i>M</i>	<i>SD</i>	Range	<i>n</i>	<i>M</i>	<i>SD</i>	Range	<i>n</i>
34.895	13	4.4	10–18	3	24	5.8	19–31	5	23	9.9	11–34	5
38.054	16	1.6	14–17	3	29	10.9	20–44	4	36	11.3	19–49	5
40.000	22	9.5	16–33	3	47	14.3	31–61	4	49	10.8	36–62	5
42.000	34	6.4	30–41	3	57	11.9	47–70	3	63	5.0	59–66	2
43.000	38	7.2	32–46	3	50	—	—	1	63	—	—	1
45.000	60	5.5	54–65	3	>72	—	—	0	>72	—	—	0

0.250 kHz can be attributed to the sound-attenuating characteristics of our single-walled booth relative to those used by Pfingst et al. (1978) and Stebbins et al. (1966).

Vocal Behavior, Socioecology, and Morphology

The finding that humans can exhibit greater sensitivity than the monkey species at some low frequencies is consistent with both an observed general correlation between low-frequency sensitivity and a lack of high-frequency acuity (Masterton et al., 1969) and with previous evidence from Old World primates (Behar et al., 1965; Pfingst et al., 1975). However, we found no significant differences between the monkey species in the region of low frequency. Contrary to our specific expectation that de Brazza's monkeys might resemble blue monkeys in exhibiting enhanced low-frequency sensitivity because of their common use of boom calls (Brown & Waser, 1984), the former were no more sensitive than vervets. In Figure 6 these normalized data clearly show that blue monkeys stand alone among the species tested in having low-frequency hearing superior to that of humans. Our de Brazza's monkeys appeared to be somewhat more sensitive than macaques but did not approach blue monkeys.

The lack of enhanced low-frequency sensitivity in de Brazza's monkeys may be related to differences in the acoustic characteristics of the respective habitats of this species and

blue monkeys. Loud vocalizations like boom calls have been proposed to function in long-distance mediation of intergroup spacing (Aldrich-Blake, 1970; Gautier, 1975; Struhsaker, 1969). Brown and Waser's (1984; Waser & Brown, 1984) report of a "sound window" for low-frequency signals in the tropical rain forest habitat of blue monkeys is consonant with this proposal. In the rain forest there is substantially less ambient background noise between 100 and 1000 Hz than at other frequencies and a minimal propagation-related amplitude decrement for sounds between 125 and 200 Hz. Thus, low-pitched sounds are readily detectable over long distances. Subsequent tests in a riverine forest, the natural habitat of de Brazza's monkeys, failed to show a comparable low-frequency window (Waser & Brown, 1986).

These findings suggested the hypothesis that de Brazza's monkeys exhibit enhanced sensitivity to low-frequency sounds presented in noise (the masked threshold). However, other evidence indicated that de Brazza's monkeys may simply not be using boom calls for long-distance localization of neighboring groups and therefore have little need for acute sensitivity to low-pitched sounds. Gautier-Hion and Gautier (1978) observed that de Brazza's monkey boom calls are very difficult to localize in their riverine environment, even within 180°. Furthermore, they described the boom as occurring concomitantly with branch shaking, a potentially salient visual signal. Both of these possible explanations of low-fre-

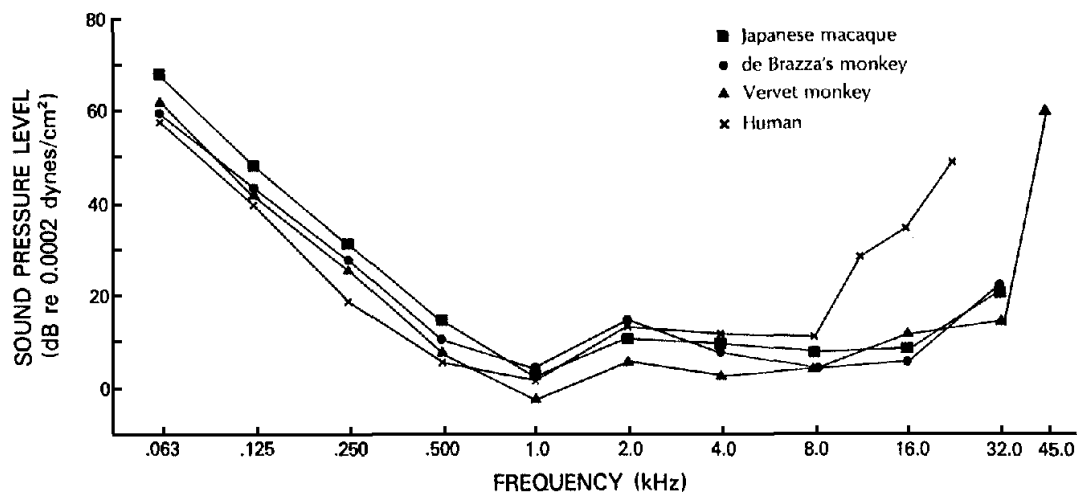


Figure 2. Mean audiograms for the three monkey species and the two human subjects most sensitive to high frequencies.

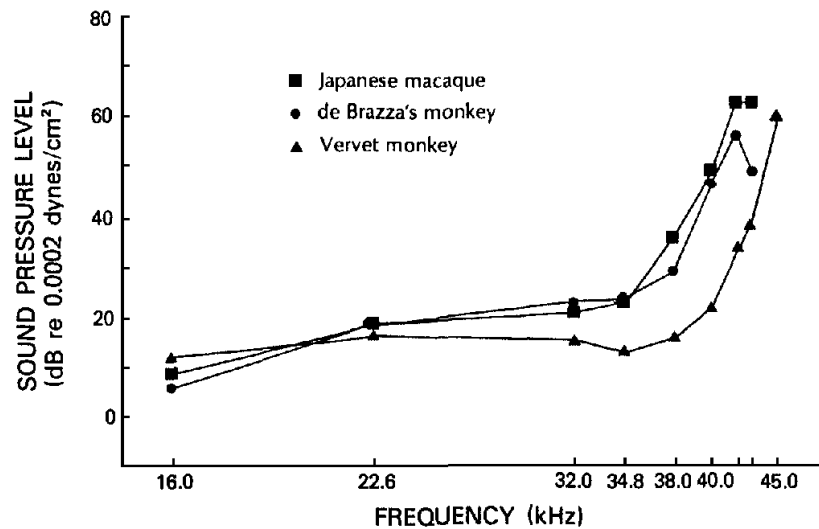


Figure 3. Mean audiograms for each monkey species at the highest frequencies tested.

quency hearing differences between blue monkeys and de Brazza's monkeys are indicative of significantly different and habitat-specific evolutionary pressures that may have led to divergence in the use and perception of a possibly homologous call.

It is more difficult to speculate about the apparently greater sensitivity of vervets in the middle frequency range. Waser and Brown (1986) suggested that the acoustics of the vervet's savanna habitat should encourage elaboration and variation of short-range calls because of attenuation-related constraints on long-distance signaling. Ambient noise characteristics that they reported for this environment may further place a premium on sensitivity in the 0.5 to 5.0 kHz region. Vervets do in fact have a diverse repertoire of short-range sounds, many of which concentrate substantial energy in the middle frequencies and exhibit quite subtle, distinctive acoustic features

(Owren, 1986; Seyfarth & Cheney, 1984; Struhsaker, 1967). However, Japanese macaques, which inhabit evergreen and deciduous forests (Eaton, 1976; Takasaki, 1981), are also known to use a complex vocal repertoire with fine structural variation in this frequency range (Green, 1975). Yet our Japanese animals were no more sensitive than the de Brazza's monkeys.

The finding that vervets clearly exhibit the best hearing above 32.0 kHz is consistent with Masterton et al.'s (1969) observation of an inverse relation between mammalian head sizes and hearing ranges. Our demonstration of this effect in nonhuman primates reinforces R. S. Heffner and Heffner's (1982) suggestion that humans need not be considered aberrant in their relative lack of high-frequency sensitivity. Rather than being a special adaptation related to speech communication, for instance, the restricted range of human hearing

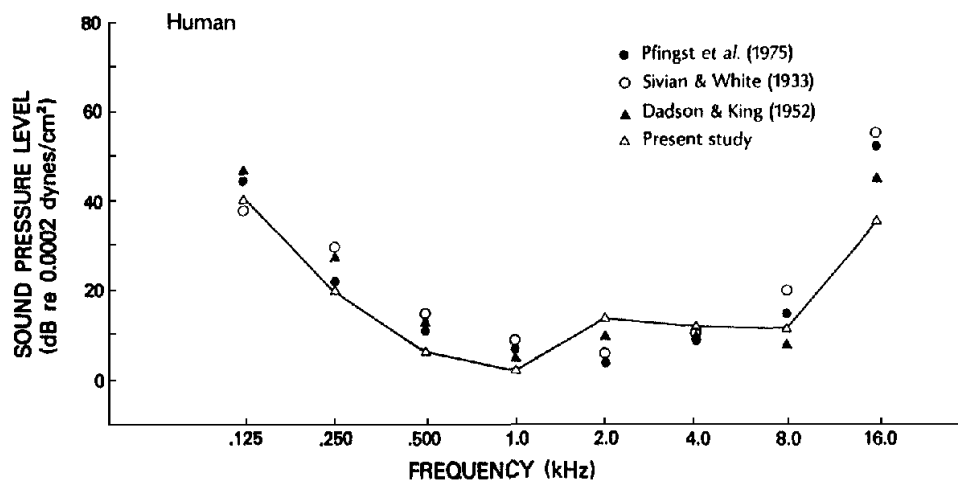


Figure 4. Mean audiograms for the two human subjects in the present study with the best high-frequency hearing and those for several previous investigations of human hearing.

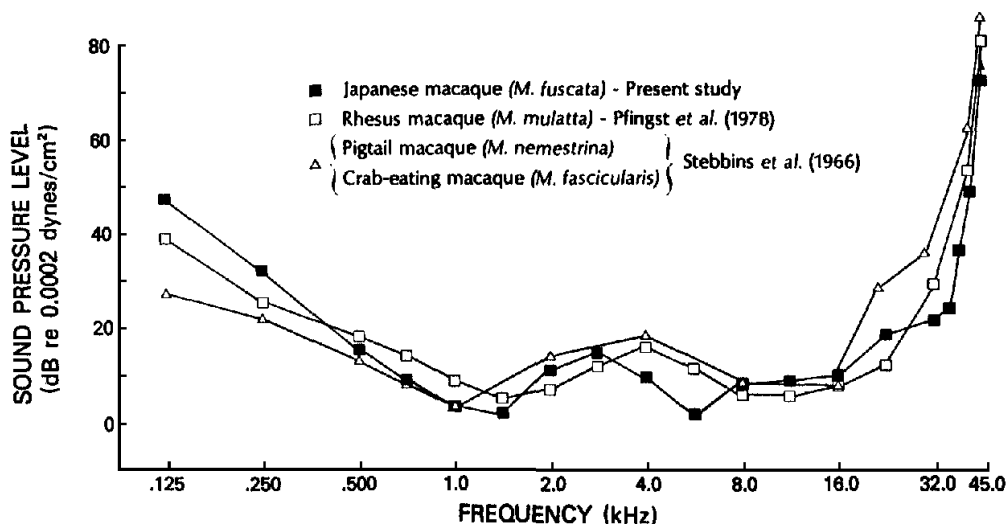


Figure 5. The mean audiogram for Japanese macaque subjects in the present study and several other macaque species tested in previous investigations.

may simply be a consequence of size-related selective pressures operating both on mammals generally and within the primate order specifically.

Evidence that vervets use acoustic energy in the upper ranges of their hearing for localization is still lacking, however. Reviewing the available evidence, Gourevitch (1978) questioned the efficacy of high-frequency cues for accurate sound localization by nonhuman primates. For example, rhesus and pigtailed macaques are able to localize pure-tone stimuli by means of binaural differences in frequency-intensity spectra, but localization at frequencies above 1 kHz is highly variable across frequencies and between individual subjects (Brown,

Beecher, Moody, & Stebbins, 1978). Better localization is achieved with broadband noise stimuli centered at various frequencies up to 16 kHz, but this result may indicate the operation of a mechanism sensitive to interaural differences in envelope fluctuations over time, rather than frequency-intensity shadowing (Brown, Beecher, Moody, & Stebbins, 1980). However, evidence that is directly relevant to the very high frequency range that we examined is not currently available.

In conjunction with other available data, our results clearly indicate the existence of species-typical variation in the absolute auditory sensitivities of Old World monkeys. Broadly

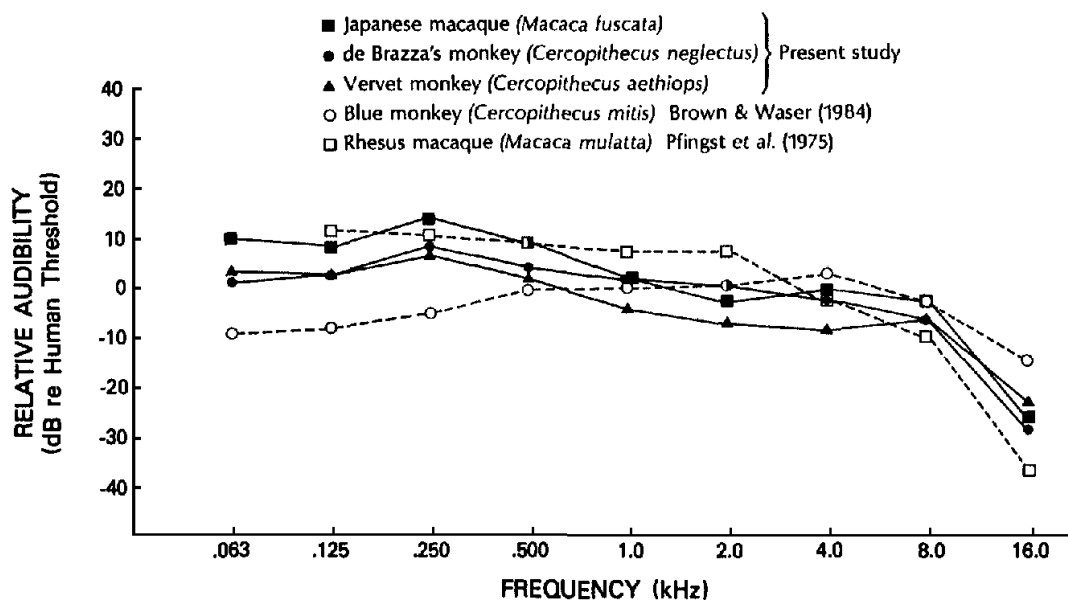


Figure 6. Audibility functions for subjects in the present and two previous studies, normalized with respect to the human controls tested in each study.

based theories of the evolution of hearing, such as that proposed by Masterton et al. (1969), and detailed study of each species's characteristic socioecology and auditory communication, as exemplified by the work of Brown and Waser (1984; Waser & Brown, 1984, 1986), will both be necessary in understanding this variation.

References

- Aldrich-Blake, F. P. G. (1970). *The ecology and behaviour of the blue monkey, Cercopithecus mitis stuhlmanni*. Unpublished doctoral dissertation, University of Bristol, Bristol, England.
- Behar, I., Cronholm, J. N., & Loeb, M. (1965). Auditory sensitivity of the rhesus monkey. *Journal of Comparative and Physiological Psychology*, 59, 426-428.
- Brown, C. H., Beecher, M. D., Moody, D. B., & Stebbins, W. C. (1978). Localization of pure tones by Old World monkeys. *Journal of the Acoustical Society of America*, 63, 1484-1492.
- Brown, C. H., Beecher, M. D., Moody, D. B., & Stebbins, W. C. (1980). Localization of noise bands by Old World monkeys. *Journal of the Acoustical Society of America*, 68, 127-132.
- Brown, C. H., & Waser, P. M. (1984). Hearing and communication in blue monkeys (*Cercopithecus mitis*). *Animal Behaviour*, 32, 66-75.
- Dadson, R. S., & King, J. H. (1952). A determination of the normal threshold of hearing and its relation to the standardization of audiometers. *Journal of Laryngology and Otology*, 66, 366-378.
- Eaton, G. G. (1976). The social organization of Japanese macaques. *Scientific American*, 235, 96-106.
- Gautier, J.-P. (1975). *Étude comparée des systèmes d'intercommunication sonore chez quelques Cercopithecines forestiers africains—Mise en évidence de corrélations phylogénétiques et socioécologiques* [A comparative study of the vocal communication systems of some African forest cercopithecines—Producing evidence of correlations between phylogeny and socioecology]. Unpublished doctoral dissertation, Université de Rennes, Paris.
- Gautier-Hion, A., & Gautier, J.-P. (1978). Le singe de Brazza: Une stratégie originale [The de Brazza's monkey: An original strategy]. *Zeitschrift für Tierpsychologie*, 46, 84-104.
- Gourevitch, G. (1978). Auditory lateralization and localization in monkeys. In D. J. Chivers & J. Herbert (Eds.), *Advances in primatology* (Vol. 1, pp. 721-733). London: Academic Press.
- Green, S. (1975). The variation of vocal pattern with social situation in the Japanese monkey (*Macaca fuscata*): A field study. In L. Rosenblum (Ed.), *Primate behavior* (Vol. 4, pp. 1-102). New York: Academic Press.
- Heffner, H. E., & Heffner, R. S. (1984). Temporal lobe lesions and perception of species-specific vocalizations by macaques. *Science*, 226, 75-76.
- Heffner, R. S., & Heffner, H. E. (1982). Hearing in the elephant (*Elephas maximus*): Absolute sensitivity, frequency discrimination, and sound localization. *Journal of Comparative and Physiological Psychology*, 96, 926-944.
- Hienz, R. D., Turkhan, J. S., & Harris, A. H. (1982). Pure tone thresholds in the yellow baboon. *Hearing Research*, 8, 71-75.
- Hill, W. C. O. (1966). *Primates: Comparative anatomy and taxonomy: VI. Cercopithecidae*. Edinburgh, Scotland: Edinburgh University Press.
- Hill, W. C. O. (1971). *Primates: Comparative anatomy and taxonomy: VII. Cynopithecinae: Cercocebus, Macaca, Cynopithecus*. Edinburgh, Scotland: Edinburgh University Press.
- Hopp, S. L. (1985). *Differential sensitivity of Japanese monkeys (Macaca fuscata) to variations along a synthetic vocal continuum*. Unpublished doctoral dissertation, Indiana University, Bloomington.
- Killion, M. C. (1978). Revised estimate of minimum audible pressure: Where is the 'missing 6 dB'? *Journal of the Acoustical Society of America*, 63, 1501.
- Masterton, B., Heffner, H., & Ravizza, R. (1969). The evolution of human hearing. *Journal of the Acoustical Society of America*, 45, 966-985.
- Owren, M. J. (1986). *The acoustic features of alarm calls in vervet monkeys (Cercopithecus aethiops): Perspectives on primate vocal behavior and the evolution of human speech*. Unpublished doctoral dissertation, Indiana University, Bloomington.
- Petersen, M. R. (1982). The perception of species-specific vocalizations by primates: A conceptual framework. In C. T. Snowdon, C. H. Brown, & M. R. Petersen (Eds.), *Primate communication* (pp. 171-211). New York: Cambridge University Press.
- Petersen, M. R., Zoloth, S. R., Beecher, M. D., Green, S., Marler, P. R., Moody, D. B., & Stebbins, W. C. (1984). Neural lateralization of vocalizations by Japanese macaques: Communicative significance is more important than acoustic structure. *Behavioral Neuroscience*, 98, 779-790.
- Pfingst, B., Hienz, R., & Miller, J. (1975). Reaction time procedure for measurement of hearing: II. Threshold functions. *Journal of the Acoustical Society of America*, 57, 431-436.
- Pfingst, B., Laycock, J., Flammino, F., & Lonsbury-Martin, B. (1978). Pure-tone thresholds for the rhesus monkey. *Hearing Research*, 1, 43-47.
- Rudmose, W. (1982). The case of the missing 6 dB. *Journal of the Acoustical Society of America*, 71, 650-659.
- Seyfarth, R. M., & Cheney, D. L. (1984). The acoustic features of vervet monkey grunts. *Journal of the Acoustical Society of America*, 75, 1623-1628.
- Sinnot, J. M., & Adams, F. (1987). Differences in sensitivity of humans and monkeys to cues underlying voice-onset-time. *Journal of the Acoustical Society of America*, 82, 1539-1547.
- Sinnot, J. M., Owren, M. J., & Petersen, M. R. (1987a). Auditory duration discrimination in Old World monkeys (*Macaca, Cercopithecus*). *Journal of the Acoustical Society of America*, 82, 465-470.
- Sinnot, J. M., Owren, M. J., & Petersen, M. R. (1987b). Auditory frequency discrimination in primates: Species differences (*Cercopithecus, Macaca, Homo*). *Journal of Comparative Psychology*, 101, 126-131.
- Sinnot, J. M., Petersen, M. R., & Hopp, S. L. (1985). Frequency and intensity discrimination in humans and monkeys. *Journal of the Acoustical Society of America*, 78, 1977-1985.
- Sivian, L. J., & White, S. D. (1933). On minimum audible fields. *Journal of the Acoustical Society of America*, 4, 288-321.
- Stebbins, W. C. (1973). Hearing of Old World monkeys (*Cercopithecinae*). *American Journal of Physical Anthropology*, 38, 357-364.
- Stebbins, W. C., Green, S., & Miller, F. (1966). Auditory sensitivity of the monkey. *Science*, 153, 1646-1647.
- Struhsaker, T. (1967). Auditory communication among vervet monkeys (*Cercopithecus aethiops*). In S. A. Altmann (Ed.), *Social communication among primates* (pp. 281-324). Chicago: University of Chicago Press.
- Struhsaker, T. (1969). Correlates of ecology and social organization among African cercopithecines. *Folia Primatologica*, 11, 8-118.
- Takasaki, H. (1981). Troop size, habitat quality, and home range area in Japanese macaques. *Behavioral Ecology and Sociobiology*, 9, 277-281.
- Waser, P. M., & Brown, C. H. (1984). Is there a "sound window" for primate communication? *Behavioral Ecology and Sociobiology*, 15, 73-76.
- Waser, P. M., & Brown, C. H. (1986). Habitat acoustics and primate communication. *American Journal of Primatology*, 10, 135-154.

Appendix

Signal Calibration

The miniature electret microphone was first calibrated with respect to the ½-in. condenser microphone–6-cc coupler system. This process entailed attaching the former to the latter in a free-field setting inside the testing booth and noting the electret's response at voltage levels that produced a 90-dB output from the ½-in. microphone at frequencies between 0.063 and 4.0 kHz. These voltages were then used to produce signals applied to the electret microphone when it was placed at the auditory canal openings of three monkeys and one human, and any differences from previous response levels were noted. The normalized electret microphone outputs are shown for each frequency in Table A-1 relative to the 90-dB standard. Values deviating from 90 dB indicate changes in signal strength because of differences between the ear volume being enclosed by the standard earphone and the volume of the earphone coupler. No differences were noted for the human, but differences were found at all frequencies below 4.0 kHz for the monkeys.

Received February 20, 1987
Revision received January 30, 1988
Accepted January 30, 1988 ■

Table A-1

Calibration Results for the Miniature Electret Condenser Microphone Relative to the ½-in. Condenser Microphone–6-cc Coupler System

Frequency (in kilohertz)	Monkeys	Humans
0.063	80	89
0.088	78	90
0.125	80	90
0.176	81	91
0.250	79	89
0.353	78	90
0.500	75	90
0.707	75	90
1.000	77	89
1.414	80	89
2.000	82	90
2.828	87	88
4.000	90	90

Call for Nominations for Editor of *Behavioral Neuroscience*

The Publications and Communications Board has opened nominations for the editorship of *Behavioral Neuroscience* for the years 1990–1995. Richard F. Thompson is the incumbent editor. Candidates must be members of APA and should be available to start receiving manuscripts in early 1989 to prepare for issues published in 1990. Please note that the P&C Board encourages more participation by women and ethnic minority men and women in the publication process and would particularly welcome such nominees. Submit nominations no later than August 1, 1988 to

Martha Storandt
Department of Psychology
Washington University
St. Louis, Missouri 63130.

Other members of the search committee are Byron Campbell, Mortimer Mishkin, Mark Rosenzweig, and Shepard Siegel.
